

DUAL-BRANCH SPIRAL INTERSECT NETWORK FOR MULTIMODAL SENTIMENT ANALYSIS

Hongfei Gao¹, Xinhua Zhu^{2,*}, Kunhao Ma¹

¹School of Computer Science and Engineering

²Key Lab of Education Blockchain and Intelligent Technology Ministry of Education
Guangxi Normal University
Guilin 541004, China

ABSTRACT

Multimodal Sentiment Analysis (MSA) predicts human emotions and opinions through multiple heterogeneous modalities. Current state-of-the-art models alleviate ambiguity and noise in non-text modalities and enhance their emotional representations via text-dominated cross-modal interaction. However, this process also causes them to lose part of their unique information, limiting the accuracy of emotional intensity prediction. To address this issue, we propose a Dual-branch Spiral Intersect Network (DSIN), which consists of a Multimodal Interaction Branch (MIB) and a Unimodal Preservation Branch (UPB). The MIB achieves unimodal feature enhancement and cross-modal information complementarity. The UPB preserves unimodal unique information through parameter sharing. The framework also adopts a hierarchical architecture to capture deep emotional cues. Experimental results on the CMU-MOSI and CMU-MOSEI datasets demonstrate the effectiveness of the proposed DSIN. We release the code at: <https://github.com/Reedfoley/DSIN>.

Index Terms— Multimodal sentiment analysis, Multimodal Representation Learning, Cross-modal interactions

1. INTRODUCTION

In recent years, Multimodal Sentiment Analysis (MSA) mines and analyzes human emotions and opinions based on three heterogeneous modalities contained in videos: text, audio, and vision [1, 2]. It has been widely applied in fields such as marketing management, social media analysis, and human-computer interaction [3]. Due to the heterogeneity of multimodality, each modality can provide unique information, and they complement each other, making sentiment analysis more comprehensive and accurate [4].

Early studies mostly adopted basic fusion networks to capture multimodal emotional information [5, 6]. This strategy cannot fully explore the unique value of each

modality, resulting in a bottleneck in prediction performance. With the deepening of research, researchers have found that the text modality plays a key role in MSA [7, 8, 9]. For example, HDRD [10] uses scaling factors to integrate the rich emotional information of the text modality into the video and audio modalities, while TCHFN [11] is text-centered and uses a hierarchical fusion architecture to integrate information from the three modalities.

Meanwhile, due to the unique ability of the attention mechanism to focus on key information, researchers have begun to use it in the process of modal interaction to capture intra-modal and inter-modal emotional information [12, 13]. For instance, SSLMM [14] utilizes multi-head graph attention to aggregate information of neighboring nodes within the same modality and refine intra-modal features. MKGN [15] employs bidirectional cross-attention to enhance multimodal feature interaction.

However, these aforementioned methods still have limitations. Although inter-modal information interaction can supplement the emotional information of each modality, under the strong guidance of the text modality, the interacted non-text modalities tend to shift towards the text modality in terms of emotional polarity and intensity, losing their own unique information and limiting the accuracy of emotion prediction.

To address this issue, we propose a Dual-branch Spiral Intersect Network (DSIN), aiming to preserve the unique discriminative information of each modality during multimodal interaction. Specifically, DSIN consists of a Multimodal Interaction Branch (MIB) and a Unimodal Preservation Branch (UPB), both of which perform feature processing through a hierarchical encoder structure, where each layer includes an intra-modal self-attention encoder (Intra-MSAE) and an inter-modal cross-attention encoder (Inter-MCAE). The MIB branch focuses on cross-modal information interaction. In each layer, it first enhances unimodal features through the Intra-MSAE to extract fine-grained emotional cues, and then realizes dynamic transfer and complementarity of inter-modal emotional information through the Inter-MCAE, thereby enhancing the semantic representation capability of each modality. The UPB focuses on unimodal information fidelity, which only purifies the original unique emotional information

*Corresponding author. Email: zhx429@263.net. This work was supported in part by the National Natural Science Foundation of China under the contract numbers 62062012.

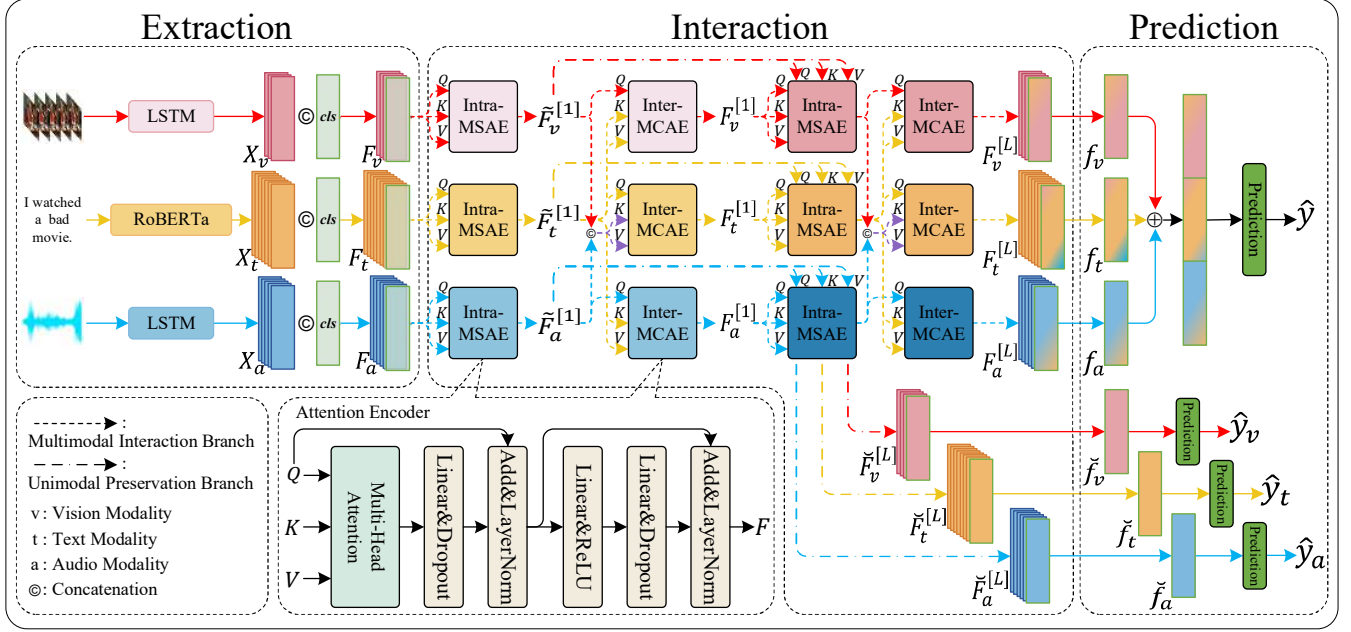


Fig. 1. The overall framework of DSIN.

of each modality through the Intra-MSAE while isolating cross-modal interference to ensure its independence and integrity. Experimental results show that DSIN achieves state-of-the-art performance on the CMU-MOSI [16] and CMU-MOSEI [17] datasets, and extensive ablation experiments verify the effectiveness and superiority of our core components.

2. METHODOLOGY

The overall framework of DSIN is shown in Figure 1. DSIN first uses RoBERTa [18] on the text modality to obtain text feature $X_t \in \mathbb{R}^{l_t \times d}$, and processes the audio and visual modalities via LSTM to generate audio feature $X_a \in \mathbb{R}^{l_a \times d}$ and visual feature $X_v \in \mathbb{R}^{l_v \times d}$, where $l_m \in \{t, a, v\}$ denotes the sequence length of each modality, and d denotes the dimension of the feature. We draw on the design idea of MTAMW [19] and concatenate a learnable $cls \in \mathbb{R}^{1 \times d}$ vector at the starting position of the X_t , X_a , and X_v sequences respectively, resulting in $F_t \in \mathbb{R}^{(l_t+1) \times d}$, $F_a \in \mathbb{R}^{(l_a+1) \times d}$, and $F_v \in \mathbb{R}^{(l_v+1) \times d}$. This cls vector aims to learn global semantic information through subsequent interaction processes and serve as the core basis for final emotion prediction.

2.1. Multimodal Interaction Branch

In the MIB, each modality passes through the corresponding Intra-MSAE and Inter-MCAE in each layer to complete unimodal feature enhancement and inter-modal information interaction. The processing flows corresponding to the text, vision, and audio modalities are marked

with yellow, red, and blue dashed lines respectively in the interaction part of Figure 1.

Both Intra-MSAE and Inter-MCAE adopt the attention encoder architecture. They first pass through an attention layer defined as:

$$Att(Q, K, V) = SoftMax\left(\frac{QW_Q(KW_K)^T}{\sqrt{d_k}}\right)VW_V \quad (1)$$

where Q, K, V represent the query, key, and value matrices, respectively, and $W_Q, W_K, W_V \in \mathbb{R}^{d \times d_k}$ denote learnable weight matrices. The output of the attention layer is processed through feed-forward neural network (FFN), regularization, residual connection, layer normalization, and activation function to output the encoder result.

Taking the first layer as an example, for any modal feature, it is simultaneously input into the Intra-MSAE as Q, K , and V . Through the self-attention mechanism, fine-grained emotional cues are captured to achieve the enhancement of unimodal features. The process is expressed as:

$$\tilde{F}_m^{[1]} = Intra-MSAE(Att(F_m, F_m, F_m)) \quad (2)$$

where $m \in \{t, a, v\}$. Then, considering that the text modality is the primary contributing modality in MSA [20], we adopt cross-attention to design two types of directional interactions to balance the contribution of modalities. For the audio and vision modalities, their features enhanced by Intra-MSAE serve as Q respectively, and the text enhanced features are used as K and V input into the Inter-MCAE. The audio and vision modalities only perform directional interaction with the text modality, effectively integrating the core semantic information of the

text to realize semantic calibration and enhancement of their own features. The process is expressed as:

$$F_n^{[1]} = \text{Inter-MCAE} \left(\text{Att} \left(\tilde{F}_n^{[1]}, \tilde{F}_t^{[1]}, \tilde{F}_t^{[1]} \right) \right) \quad (3)$$

where $n \in \{a, v\}$. For the text modality, its enhanced features serve as Q , and the enhanced visual and audio features are concatenated along the sequence dimension into an audio-visual joint feature, which is used as K and V input into the Inter-MCAE. Utilizing the fine-grained emotional cues carried by non-text modalities, emotional contextual information is supplemented for the text modality to improve the richness of its emotional expression. The process is expressed as:

$$\begin{aligned} \tilde{F}_{a,v}^{[1]} &= \text{Concatenation} \left[\tilde{F}_a^{[1]}, \tilde{F}_v^{[1]} \right] \\ F_t^{[1]} &= \text{Inter-MCAE} \left(\text{Att} \left(\tilde{F}_t^{[1]}, \tilde{F}_{a,v}^{[1]}, \tilde{F}_{a,v}^{[1]} \right) \right) \end{aligned} \quad (4)$$

where $\tilde{F}_{a,v}^{[1]} \in \mathbb{R}^{(l_a+l_v+2) \times d}$. After L layers, the MIB outputs $F_t^{[L]}$, $F_a^{[L]}$, and $F_v^{[L]}$ as the multimodal feature input for subsequent emotion prediction.

2.2. Unimodal Preservation Branch

In the UPB, each modality extracts and enhances intra-modal fine-grained emotional cues through only the Intra-MSAE in each layer to ensure the purity of unimodal features. Among them, the processing flows of text, visual, and audio modalities are marked with yellow, red, and blue dashed lines with dots respectively in the interaction part of Figure 1.

To achieve dual-branch collaborative optimization, the UPB shares the learnable parameters of the Intra-MSAE in each layer with the MIB. In the first layer, the input of the UPB is consistent with that of the MIB, both being F_t , F_a , and F_v , and their outputs after passing through the Intra-MSAE are also the same, both being $\tilde{F}_t^{[1]}$, $\tilde{F}_a^{[1]}$, and $\tilde{F}_v^{[1]}$. Subsequently, the input of the Intra-MSAE in the i -th layer of the UPB is the output of its $(i-1)$ -th layer, and the process is expressed as:

$$\check{F}_m^i = \text{Intra-MSAE} \left(\text{Att} \left(\check{F}_m^{i-1}, \check{F}_m^{i-1}, \check{F}_m^{i-1} \right) \right) \quad (5)$$

where $m \in \{t, a, v\}$. After L layers, the UPB outputs pure unimodal features $\check{F}_t^L$, $\check{F}_a^L$, and $\check{F}_v^L$, as the basis for judging unimodal unique information.

2.3. Multi-Task Prediction

In the prediction phase, we did not adopt the feature matrix dimensionality reduction strategy. Instead, we extracted the previously embedded *cls* vectors f_t , f_a , f_v , $\check{f}_t$, $\check{f}_a$, and $\check{f}_v$ from $F_t^{[L]}$, $F_a^{[L]}$, $F_v^{[L]}$, $\check{F}_t^L$, $\check{F}_a^L$, and $\check{F}_v^L$ output by MIB and UPB as the basis for prediction. During multi-layer iterative interaction, the *cls* vectors aggregated key emotional semantic information of various features through the

attention mechanism, ensuring prediction accuracy while avoiding the computational overhead caused by matrix dimensionality reduction.

Based on existing research [6, 7, 20], we introduce a multi-task learning framework to improve the prediction performance of the model. Specifically, we concatenate f_a , f_t , and f_v in sequence along the feature dimension, and generate the multimodal sentiment prediction $\hat{y}$ through FFN. $\check{f}_t$, $\check{f}_a$, and $\check{f}_v$ are processed by independent FFNs respectively to generate unimodal sentiment predictions $\hat{y}_t$, $\hat{y}_a$, and $\hat{y}_v$. The total training loss of the model consists of four components, which is specifically expressed as follows:

$$\begin{aligned} \mathcal{L}_{main} &= \frac{1}{N} \sum_{j=1}^N \left| y^{(j)} - \hat{y}^{(j)} \right| \\ \mathcal{L}_m &= \frac{1}{N} \sum_{j=1}^N \left| y^{(j)} - \hat{y}_m^{(j)} \right| \\ \mathcal{L} &= \mathcal{L}_{main} + \mathcal{L}_t + \mathcal{L}_a + \mathcal{L}_v \end{aligned} \quad (6)$$

where $m \in \{t, a, v\}$, $y^{(j)}$ represents the label of the j -th sample, and N represents the number of samples in a batch.

3. EXPERIMENT

3.1. Datasets, Metrics, and Implementation Details

We evaluated DSIN on the most widely used benchmark datasets in MSA, namely CMU-MOSI and CMU-MOSEI. CMU-MOSI consists of 2,199 opinion segments, with annotations capturing sentiment on a scale ranging from negative to positive (-3 to 3). CMU-MOSEI comprises 23,453 annotated video clips from 1,000 speakers, each with a sentiment score interval of [-3,3]. Five metrics are used to evaluate the performance of DSIN and baselines: binary accuracy (Acc2), 7-class accuracy (Acc7), F1 score (F1), Pearson correlation (Corr), and mean absolute error (MAE). The initial learning rate of the model was set to $5e-6$, and the number of interaction layers was set to 3.

3.2. Experimental Results and Analysis

To validate the performance and effectiveness of the proposed model, we compare DSIN with 10 leading models from recent two years on two benchmarks, including HDRD [10], MMML [12], DTN [21], TMT [22], TCHF [11], MCL-MCF [4], MCAN [20], DLF [13], MKGN [15], PMJRS [23], and SSLMM [14]. The results of these models are all provided by their original papers.

As shown in Table 1, the comparative results demonstrate that our proposed DSIN achieves state-of-the-art performance on all metrics across the two benchmark datasets. It is worth noting that through comparative analysis of existing methods, we have the following key findings. Compared with DTN and TMT, which focus on separating modal private emotional features, the proposed dual-branch spiral structure can more fully extract deep

Table 1. The experiment results on CMU-MOSI and CMU-MOSEI. The best results are labeled in bold.

Model	CMU-MOSI					SMU-MOSEI				
	Acc2 $\uparrow$	Acc7 $\uparrow$	F1 $\uparrow$	Corr $\uparrow$	MAE $\downarrow$	Acc2 $\uparrow$	Acc7 $\uparrow$	F1 $\uparrow$	Corr $\uparrow$	MAE $\downarrow$
HDRD	88.89	-	88.87	0.815	0.684	86.51	-	86.23	0.761	0.514
MMML	88.16	48.25	88.15	0.838	0.643	86.73	54.95	86.49	0.791	0.517
DTN	86.20	48.10	86.20	0.807	0.714	86.30	52.50	86.30	0.788	0.579
TMT	86.40	47.3	86.50	0.801	0.718	86.50	53.7	86.50	0.775	0.542
TCHFEN	86.13	44.75	86.31	0.780	0.748	86.27	53.19	86.48	0.770	0.538
MCL-MCF	87.30	-	87.20	0.799	0.692	86.40	-	86.30	0.767	0.536
MCAN	84.50	43.10	84.80	0.811	0.675	85.80	51.60	85.90	0.789	0.527
DLF	85.06	47.08	85.04	0.781	0.731	85.42	53.90	85.27	0.764	0.536
MKGN	87.10	-	86.90	0.811	0.677	88.00	-	88.00	0.806	0.520
PMJRS	88.60	43.00	88.40	-	-	88.30	55.70	87.80	-	-
SSLMM	85.10	46.80	85.40	0.798	0.703	86.00	53.30	86.10	0.756	0.551
DSIN(ours)	89.79	50.58	89.78	0.871	0.583	88.58	55.78	88.50	0.812	0.510

Table 2. Ablation study of DSIN on CMU-MOSEI.

Method	Acc2 $\uparrow$	Acc7 $\uparrow$	F1 $\uparrow$	Corr $\uparrow$	MAE $\downarrow$
DSIN(Ours)	88.58	55.78	88.50	0.812	0.510
Different Modalities					
w/o T	64.60	41.90	51.97	0.140	0.782
w/o A	85.05	49.17	85.17	0.804	0.581
w/o V	84.61	49.74	85.22	0.793	0.577
Different Losses					
w/o $\mathcal{L}_t$	87.15	53.73	86.84	0.806	0.495
w/o $\mathcal{L}_a$	87.78	51.62	87.64	0.809	0.541
w/o $\mathcal{L}_v$	87.56	52.01	87.28	0.810	0.539
Different Layers					
layers=1	87.51	55.01	87.38	0.793	0.541
layers=2	87.78	53.62	87.57	0.808	0.523
layers=3	88.58	55.78	88.50	0.812	0.510
layers=4	87.89	54.43	87.88	0.803	0.515
layers=5	88.30	51.34	88.25	0.813	0.545

unimodal unique emotional information through the UPB. Compared with HDRD and TCHFEN, which emphasize the dominant role of text, the designed asymmetric interaction mechanism achieves more balanced unimodal emotional semantic enhancement. Compared with methods such as MMML and DLF that achieve multimodal interaction based on attention mechanisms, the constructed hierarchical attention network realizes more efficient inter-modal information transfer and complementarity through multi-layer iterative interaction, enabling the accuracy of cross-modal semantic fusion to increase layer by layer with network depth. Overall, DSIN has indeed further improved the performance of MSA.

3.3. Ablation Study

We conducted extensive ablation experiments on the CMU-MOSEI dataset to investigate the impact of modal

combinations, loss function configurations, and the number of interaction layers on model performance, with experimental results shown in Table 2. First, by sequentially removing individual modalities to explore the contribution of each modality, the results confirm the dominant role of the text modality in sentiment prediction tasks. Second, by removing unimodal loss terms to verify the impact of unimodal unique information preserved by the UPB branch on prediction, the results show that the absence of unique information from any modality leads to a decrease in model performance, demonstrating the necessity of UPB for maintaining the discriminability of unimodal features. Finally, by adjusting the number of interaction layers to explore the optimization effect of the hierarchical spiral interaction mechanism, the results show that model performance reaches the optimal level when the number of interaction layers is 3.

4. CONCLUSION

In this paper, to address the key issue in multimodal sentiment analysis where the unique emotional information of non-text modalities is diluted or even lost during cross-modal interaction due to the strong dominance of the text modality, we propose the DSIN architecture. In this framework, the MIB achieves deep transfer and complementarity of cross-modal information via a hierarchical spiral interaction mechanism, while the UPB purifies and preserves the original unique emotional information of each modality through intra-modal self-attention encoders under the condition of isolating cross-modal interference. The two branches share Intra-MSAE parameters. Experimental results show that DSIN outperforms state-of-the-art methods on two mainstream multimodal sentiment analysis datasets, verifying the effectiveness of our entire framework.

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